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References Book:

1. Computer Network,Andrew S.Tanenbaum 5th Edition pearson publication
2. Computer Networking James F Kurose Pearson publication
3. Computer Networking Behrouz A.Forouzan, TMH Publication

FACULTY

HOD

05/10/24

DEAN ACADEMICS

UNIT-2

LECTURE-1

Data-linklayeruseserrorcontroltechniquestoensure thatframes,i.e.bitstreamsofdata,aretransmittedfromthesourcetothe destinationwithacertainextentof accuracy.

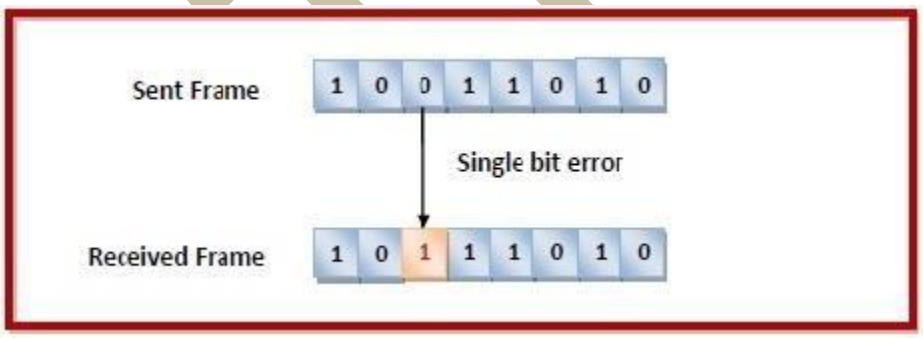
Errors

When bits are transmitted over the computer network, they are subject to get corrupted due to interferenceand network problems. The corrupted bits leads to spurious data being received by the destination and arecallederrors.

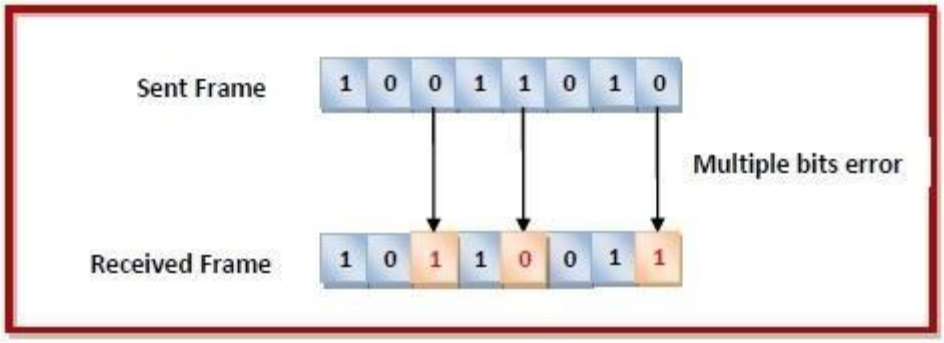
Types ofErrors

Errorscanbeofthreetypes,namelysinglebiterrors,multiplebiterrors,andbursterrors.

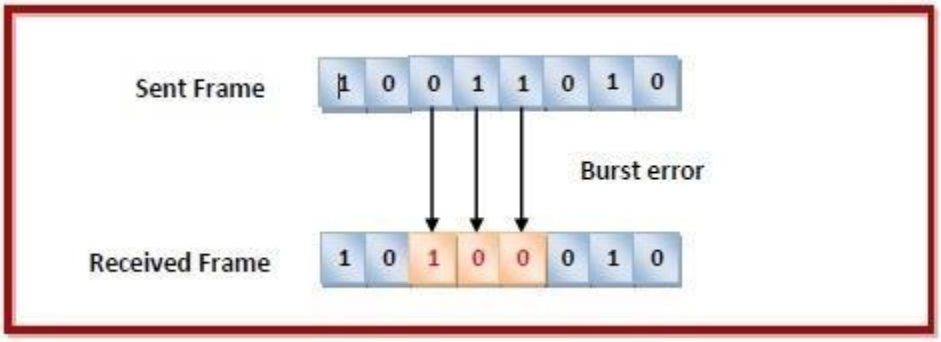
- **Singlebiterror** − Inthereceivedframe,onlyonebithas beencorrupted,i.e.eitherchangedfrom 0to1or from1to0.



- **Multiplebits error**− Inthereceivedframe,morethan onebits arecorrupted.



- **Bursterror** −Inthereceivedframe,morethan oneconsecutivebits arecorrupted.



ErrorControl

Errorcontrolcanbedoneintwoways

- **Errordetection**−Errordetectioninvolvescheckingwhetheranyerrorhasoccurredornot.Then umberoferrorbitsandthetypeoferrordoesnotmatter.
- **Errorcorrection**−Errorcorrectioninvolvesascertainingtheexactnumberofbitsthathasbeenc orruptedandthe locationof the corruptedbits.

For both error detection and error correction, the sender needs to send some additional bits along with thedata bits. The receiver performs necessary checks based upon the additional redundant bits. If it finds thatthedataisfreefromerrors,itremovestheredundantbits beforepassingthemessagetotheupperlayers.

ErrorDetectionTechniques

There are three main techniques for detecting errors in frames: Parity Check, Checksum and CyclicRedundancyCheck(CRC).

ParityCheck

The parity check is done by adding an extra bit, called parity bit to the data to make a number of 1s eitherevenincaseof evenparityoroddincaseof oddparity.

While creating a frame, the sender counts the number of 1s in it and adds the parity bit in the followingway

- Incaseofevenparity:Ifanumberof1siseventhenparitybitvalueis0.Ifthenumberof1sisoddthen paritybitvalueis1.
- In caseofodd parity:Ifanumberof1sisodd thenparity bitvalueis0.Ifanumberof1siseventhenparitybitvalueis1.

MITRC

On receiving a frame, the receiver counts the number of 1s in it. In case of even parity check, if the countof 1s iseven, the frame is accepted, otherwise, itis rejected. A similar rule is adopted for odd paritycheck.

Theparitycheckis suitableforsinglebiterrordetectiononly.

Checksum

Inthiserrordetectionscheme,thefollowingprocedureis applied

- Dataisdividedintofixedsizedframesorsegments.
- Thesenderaddsthesegmentsusing1'scomplementarithmetictogetthesum.Itthencomplement sthesumtogetthechecksumandsendsitalongwiththe dataframes.
- Thereceiveraddstheincomingsegmentsalongwiththechecksumusing1'scomplementarithme tictogetthesumandthencomplementsit.
- Iftheresult iszero,thereceivedframesareaccepted; otherwise,theyarediscarded.

CyclicRedundancyCheck(CRC)

Cyclic Redundancy Check (CRC) involves binary division of the data bits being sent by a predetermineddivisoragreeduponbythecommunicatingsystem.Thedivisorisgeneratedusingpolyno mials.

- Here, the sender performs binary division of the data segment by the divisor. It then appends theremainder called CRC bits to the end of the data segment. This makes the resulting data unitexactlydivisiblebythe divisor.
- Thereceiverdividestheincomingdataunitbythedivisor.Ifthereisnoremainder,thedataunitis assumed to be correct and is accepted. Otherwise, it is understood that the data is corrupted andisthereforerejected.

ErrorCorrectionTechniques

Error correction techniques find out the exact number of bits that have been corrupted and as well as theirlocations.Therearetwoprinciple ways

- **BackwardErrorCorrection(Retransmission)** −Ifthereceiverdetectsanerrorintheincoming frame, it requests the sender to retransmit the frame. It is a relatively simple technique.But it can be efficiently used only where retransmitting is not expensive as in fiber optics and thetimeforretransmissionislowrelativetotherequirementsoftheapplication.
- **ForwardErrorCorrection** −If the receiverdetectssome errorin the incoming

frame,itexecutes error-correcting code that generates the actual frame. This saves bandwidth required forretransmission. It is inevitable in real-time systems. However, if there are too many errors, theframesneedtoberetransmitted.

Thefourmain errorcorrectioncodes are

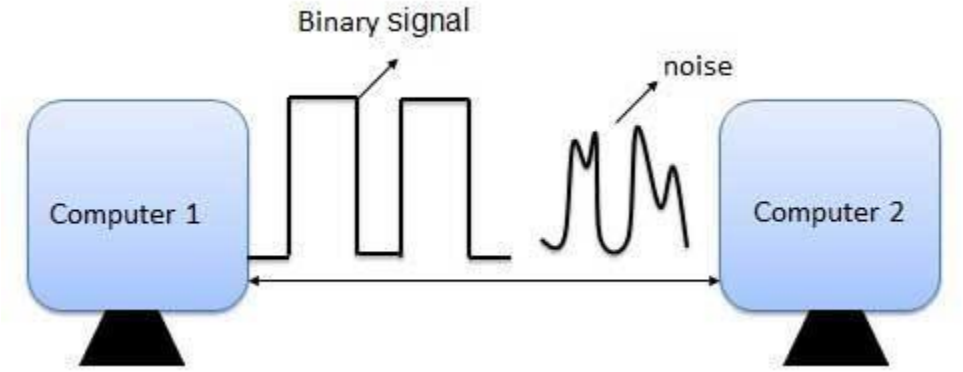
- HammingCodes
- BinaryConvolutionCode
- Reed–SolomonCode
- Low-DensityParity-CheckCode

DETECTIONVERSESCORRECTION

LECTURE 2

Whatis Error?

Error is a condition when the output information does not match with the input information. Duringtransmission, digital signals suffer from noise that can introduce errors in the binary bits travelling fromonesystemtoother.Thatmeansa0bitmaychangeto1or a1bitmaychangeto0.



Error-Detectingcodes

Whenever a message is transmitted, it may get scrambled by noise or data may get corrupted. To avoidthis, we use error-detecting codes which are additional data added to a given digital message to help usdetect if an error occurred during transmission of the message. A simple example of error-detecting codeis**paritycheck**.

Error-Correctingcodes

Along with error-detecting code, we can also pass some data to figure out the original message from thecorrupt message that we received. This type of code is called an error-correcting code. Error-correctingcodes also deploy the same strategy as error-detecting codes but additionally, such codes also detect theexactlocationof the corruptbit.

In error-correcting codes, parity check has a simple way to detect errors along with a sophisticatedmechanism to determine the corrupt bit location.Once the corrupt bit is located,its value is reverted(from0to1or1to0) togettheoriginalmessage.

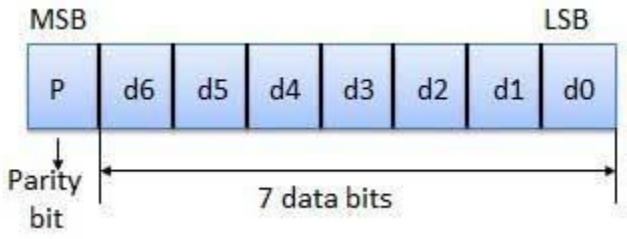
HowtoDetectandCorrectErrors?

Todetectandcorrecttheerrors,additionalbits areaddedtothedatabitsatthetimeoftransmission.

- Theadditionalbits arecalled **parity bits**.Theyallowdetectionorcorrectionoftheerrors.
- Thedatabitsalongwiththeparitybitsforma**codeword**.

ParityCheckingofErrorDetection

It is the simplest technique for detecting and correcting errors. The MSB of an 8-bits word is used as theparity bit and the remaining 7 bits are used as data or message bits. The parity of 8-bits transmitted wordcanbe eitherevenparityoroddparity.



Even parity -- Even parity means the number of 1's in the given word including the parity bit should beeven(2,4,6,).

Oddparity -- Odd parity means the numberof 1's in the given word including the parity bit should beodd(1,3,5,).

UseofParityBit

Theparitybitcanbesetto0and1dependingonthetype of theparityrequired.

- Forevenparity,thisbitissetto1or0suchthattheno.of"1bits"intheentirewordiseven.Showninfig.(a).
- Foroddparity,thisbitissetto1or0suchthattheno.of"1bits"intheentirewordisodd.Showninfig.(b).

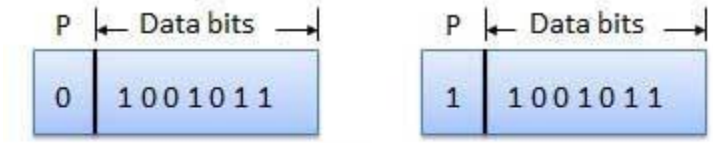


Fig. (a)

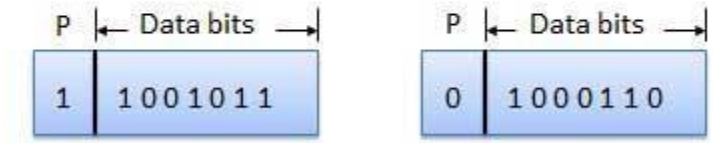
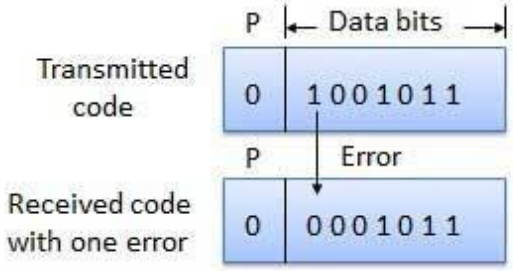


Fig. (b)

HowDoes ErrorDetectionTakePlace?

Parity checking at the receiver can detect the presence of an error if the parity of the receiver signal isdifferent from the expected parity. That means, if it is known that the parity of the transmitted signal isalways going to be "even" and if the received signal has an odd parity, then the receiver can conclude thatthereceivedsignalisnotcorrect.Ifanerrorisdetected,thenthereceiverwillignorethereceivedbytean drequestforretransmissionof thesamebytetothetransmitter.

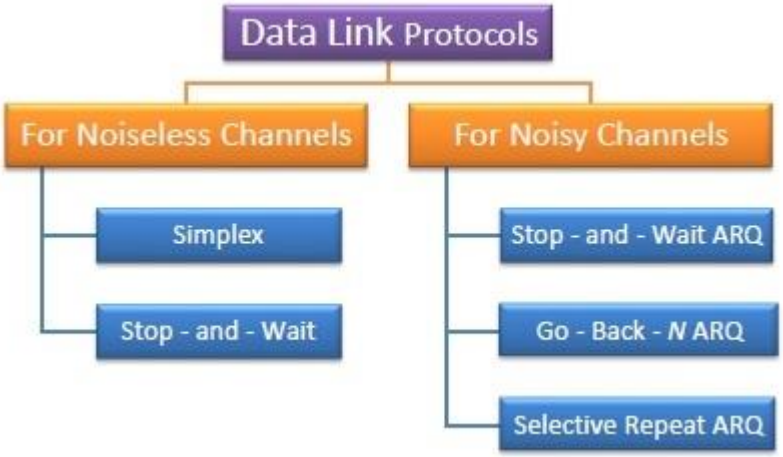


ELEMENTRY DATA LINK PROTOCOLS
LECTURE 3

Protocols in the **data link layer** are designed so that this layer can perform its basic functions: framing, error control and flow control. Framing is the process of dividing bit - streams from physical layer into data frames whose size ranges from a few hundred to a few thousand bytes. Error control mechanisms deals with transmission errors and retransmission of corrupted and lost frames. Flow control regulates speed of delivery and so that a fast sender does not drown a slow receiver.

Types of Data Link Protocols

Data link protocols can be broadly divided into two categories, depending on whether the transmission channel is noiseless or noisy.



Simplex Protocol

The Simplex protocol is hypothetical protocol designed for unidirectional data transmission over an ideal channel, i.e. a channel through which transmission can never go wrong. It has distinct procedures for sender and receiver. The sender simply sends all its data available onto the channel as soon as they are available its buffer. The receiver is assumed to process all incoming data instantly. It is hypothetical since it does not handle flow control or error control.

Simplex stop and wait protocol for an error free channel:

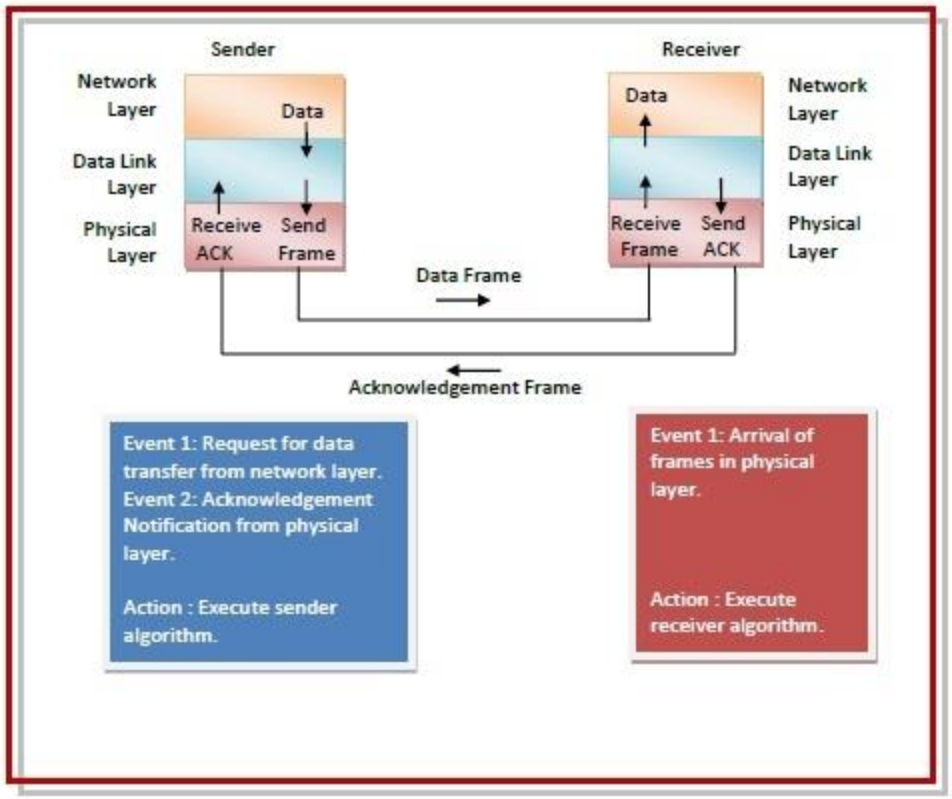
Stop – and – Wait protocol is data link layer protocol for transmission of frames over noiseless channels. It provides unidirectional data transmission with flow control facilities but without error control facilities.

This protocol takes into account the fact that the receiver has a finite processing speed. If data frames arrive at the receiver's end at a rate which is greater than its rate of processing, frames be dropped out. In order to avoid this, the receiver sends an acknowledgement for each frame

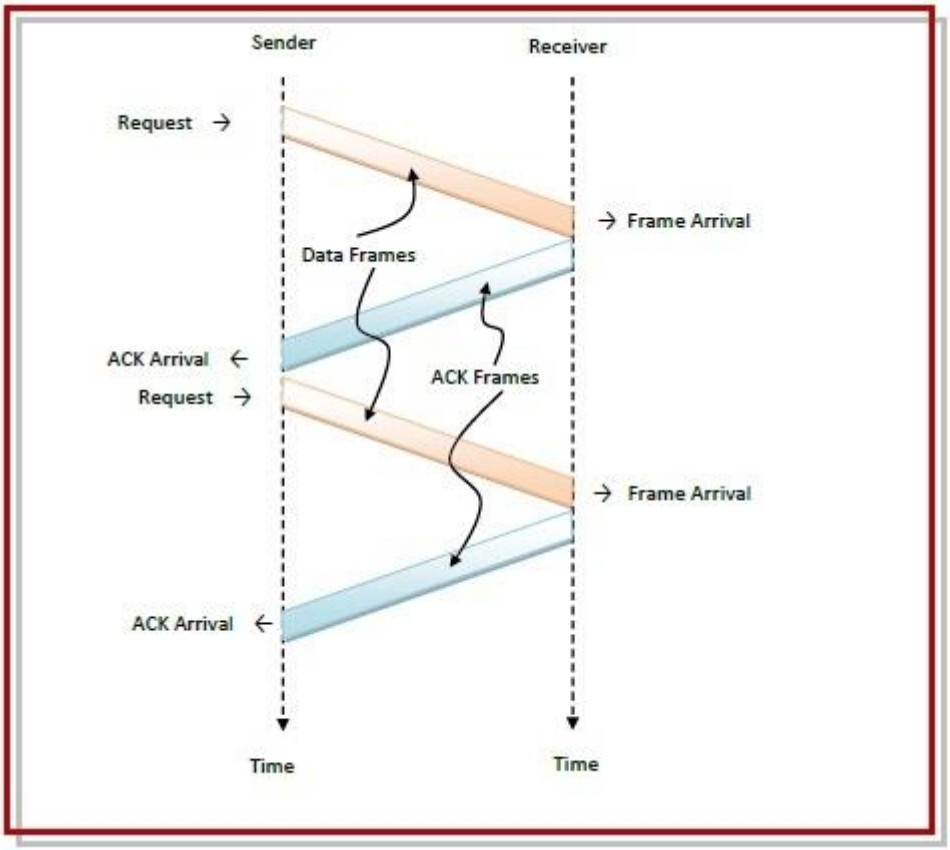
upon its arrival. The sender sends the next frame only when it has received a positive acknowledgement from the receiver that it is available for further data processing.

Design

- **Sender Site**: The data link layer in the sender site waits for the network layer for a data packet. It then checks whether it can send the frame. If it receives a positive notification from the physical layer, it makes frames out of the data and sends it. It then waits for an acknowledgement before sending the next frame.
- **Receiver Site**: The data link layer in the receiver site waits for a frame to arrive. When it arrives, the receiver processes it and delivers it to the network layer. It then sends an acknowledgement back to the sender.



The following flow diagram depicts communication via simplex stop – and – wait protocol for noiseless channel



Simplex stop and wait protocol for noisy channel:

Simplex Stop – and – Wait protocol for noisy channel is data link layer protocol for data communications with error control and flow control mechanisms. It is popularly known as Stop – and –Wait Automatic Repeat Request (Stop – and –Wait ARQ) protocol. It adds error control facilities to Stop – and – Wait protocol.

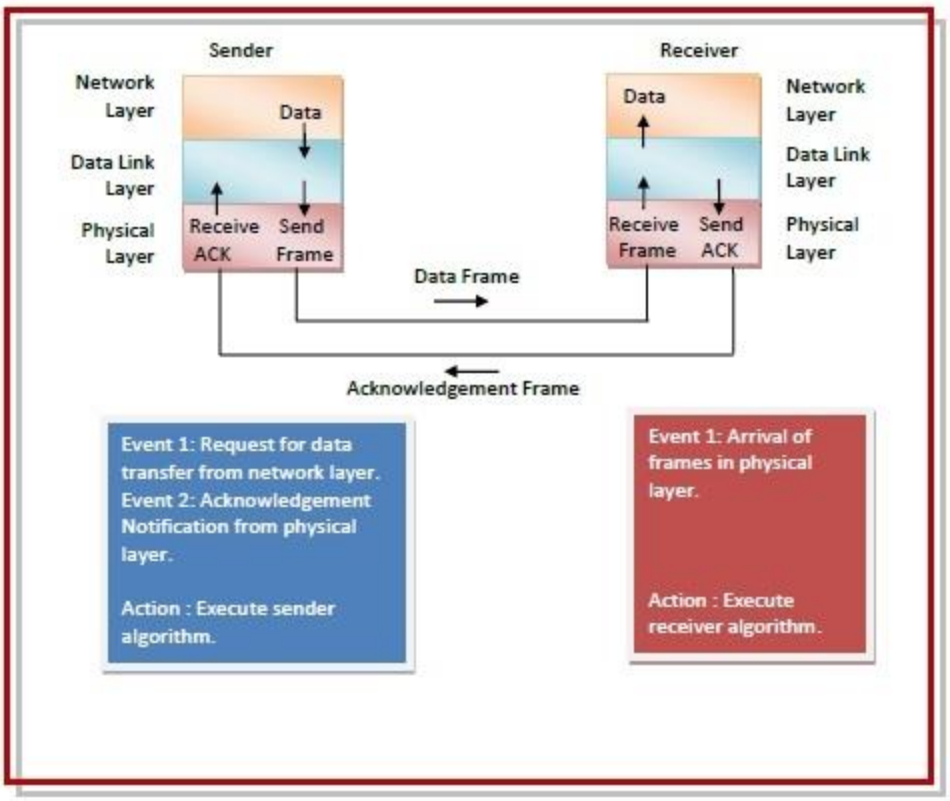
This protocol takes into account the facts that the receiver has a finite processing speed and that frames may get corrupted while transmission. If data frames arrive at the receiver's end at a rate which is greater than its rate of processing, frames can be dropped out. Also, frames may get corrupted or entirely lost when they are transmitted via network channels. So, the receiver sends an acknowledgment for each valid frame that it receives. The sender sends the next frame only when it has received a positive acknowledgment from the receiver that it is available for further data processing. Otherwise, it waits for a certain amount of time and then resends the frame.

Design

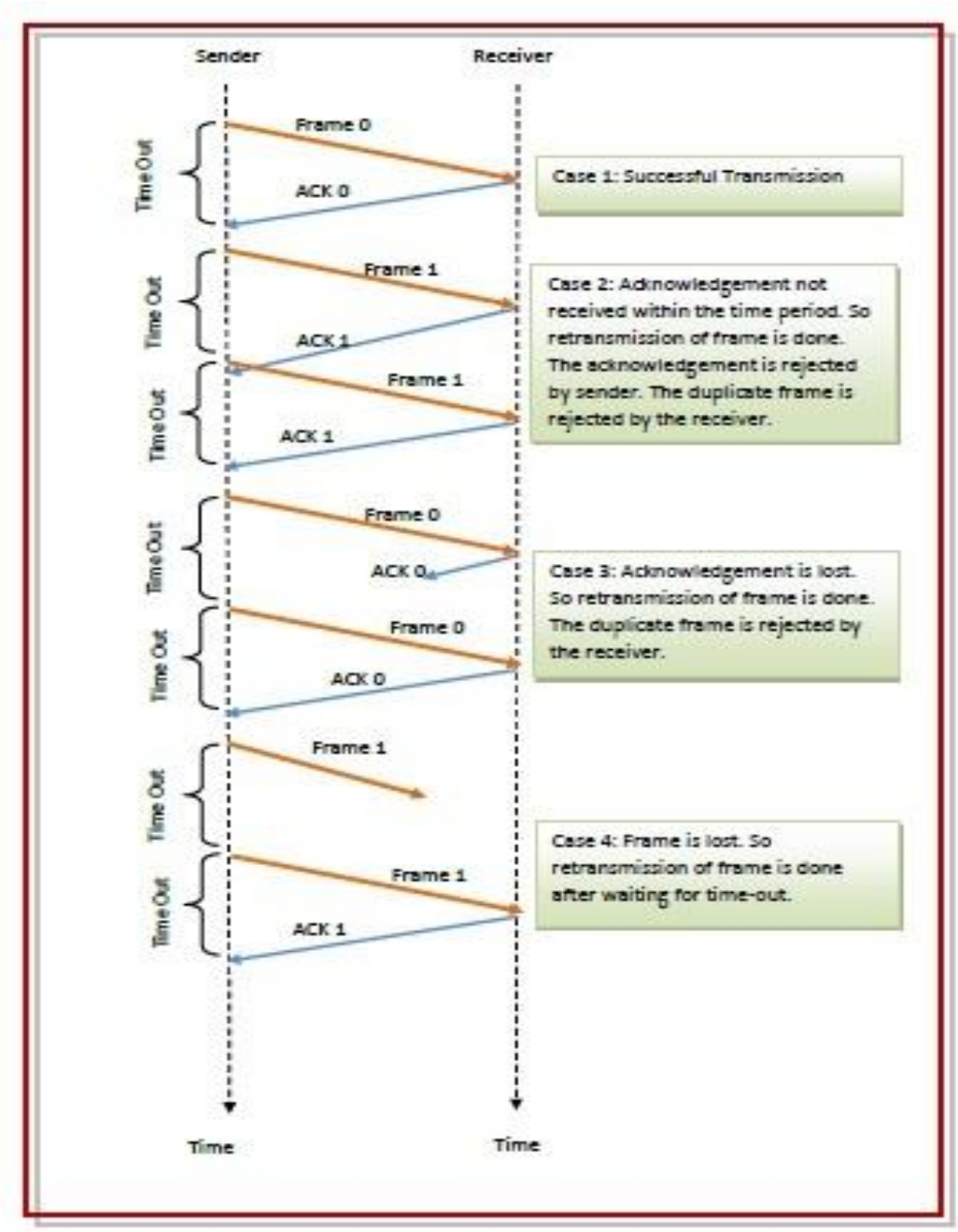
- **Sender Site** − At the sender site, a field is added to the frame to hold a sequence number. If data is available, the data link layer makes a frame with the certain sequence number and sends it. The sender then waits for arrival of acknowledgment

for a certain amount of time. If it receives a positive acknowledgment for the frame with that sequence number within the stipulated time, it sends the frame with next sequence number. Otherwise, it resends the same frame.

- **Receiver Site** − The receiver also keeps a sequence number of the frames expected for arrival. When a frame arrives, the receiver processes it and checks whether it is valid or not. If it is valid and its sequence number matches the sequence number of the expected frame, it extracts the data and delivers it to the network layer. It then sends an acknowledgement for that frame back to the sender along with its sequence number.



The following flow diagram depicts communication via simplex stop – and – wait ARQ protocol for noisy channel −



SLIDING WINDOW PROTOCOL

LECTURE 4

The sliding window protocol is a data link layer protocol that is useful in the sequential and reliable delivery of the data frames. Using the sliding window protocol, the sender can send multiple frames at a time. The sliding window protocol uses a mechanism of sequence numbers. The sender associates a sequence number to the data frames so that the receiver can use this sequence number to arrange the frames in order if any frame was re-transmitted. The sequence number also helps the receiver identify the loss of damaged packets. When the receiver receives the frame, it sends back an ACK (acknowledgment) to the sender. The ACK lets the sender know that a particular frame is received by the receiver correctly. There are two types of sliding window protocols namely - Go-Back-N ARQ, and Selective Repeat ARQ.

What is Sliding Window Protocol?

Before learning about the sliding window protocol, let us first learn about computer networks, the OSI model, and the data link layer (as the Sliding Window Protocol is related to the data link layer).

A computer network is a set of devices (computers) connected to exchange information and resources such as files etc.

The OSI model stands for Open Systems Interconnection model. It is a conceptual reference model that describes the entire flow of information from one computer to the other computer. The OSI model is a 7-layered model so it is also known as a 7-layered architecture model. The data link layer is one of the seven layers of the OSI model.

Now, let us discuss the data link layer briefly so that we can get a better understanding of the Selective Repeat ARQ protocol as it is one of the most widely used protocols of the data link layer to control errors.

The data link is the second layer of the OSI model which is used for the transmission of the error-free frames from one node to the other node. If the two computer nodes are on the same networks then the data link layer provides a connection between the two nodes. The main work of the data link layer is to convert the data into the form of frames (refer to the sub-section below for details about framing).

The functions provided by the data link layer are as follows:

- It provides reliable and efficient communication between two or more devices.
- **Framing**: Framing is the technique in which the data is divided into streams of bits (called frames) received from the network layer. Along with the conversion of data into frames, the data link layer adds a header and trailer to the frames. The header (present at the starting of the frame) contains the hardware's physical address of source and destination. The trailer (present at the end of the frame) contains the error detection and correction bits.

- The data link layer also maintains the flow control of data during transmission. Suppose that the rate of data transmission and data absorption varies then there is data loss, so the data link layer maintains the flow control.
- As we have seen above, the data link layer adds the error detection and correction bits at the end of the frames in the form of trailers. These bits are used to detect the errors and then retransmit the damaged or lost data to prevent any kind of duplication.
- It also maintains access control. In situations where two or more devices are connected to the same communication medium then the data link layer protocols determine the device that can transmit the data.

The flow control protocols involved in the data link layers are as follows:

Types of Sliding Window Protocols

There are two types of sliding window protocols. Let us discuss them one by one.

1. Go-Back-N ARQ

The Go-Back-N ARQ is one of the Sliding Window Protocol strategies that is used where reliable in-order delivery of the data packets is required. In the Go-Back-N ARQ, we use the concept of a timer. When the receiver receives the correct frame, it sends back an acknowledgment or ACK. Once the sender gets an ACK for a data frame, it shifts its window forward and sends the next frame. If the ACK is not received within the specified period then all the data frames starting from the lost frame are retransmitted.

In the Go-Back-N ARQ, the sender's window size is taken as N but the receiver's window size is always 1. Hence, the sender can send N data frames at a time but the receiver can receive only 1 frame at a time. The Go-Back-N ARQ is used for noisy channels or links and it manages the flow and error control between the sender and the receiver.

Important Points regarding the Go-Back-N ARQ:

- In the Go-Back-N ARQ, we retransmit all the data frames starting from the lost or damaged frames.
- The ACK has the sequence number of the frame that helps the sender to identify the lost frame.
- The sender sets a timer for each frame so whenever the timer is over and the sender has not received any acknowledgment, then the sender knows that the particular frame is either lost or damaged.
- The ACK has the sequence number of the frame that helps the sender to identify the lost frame.
- The sender's window size is N.
- The receiver's window size is 1.
- The sender waits for the period specified by the timer before sending the frame again. Hence, the Go-Back-N ARQ is a slower protocol than the Selective Repeat ARQ.
- In Go-Back-N ARQ protocol, the efficiency is N/(1+2xa) where a is ratio of propagation delay vs the transmission delay and N is number of packets sent.
- GoBackN is quite easy to implement but here lot of bandwidth is used in case of high error rate for the retransimmion of the entire window every time.

2. Selective Repeat ARQ

The selective repeat ARQ is one of the Sliding Window Protocol strategies that is used where reliable in-order delivery of the data packets is required. The selective repeat ARQ is used for noisy channels or links and it manages the flow and error control between the sender and the receiver.

In the selective repeat ARQ, we only retransmit the data frames that are damaged or lost. If any frame is lost or damaged then the receiver sends a negative acknowledgment (NACK) to the sender and if the frame is correctly received, it sends back an acknowledgment (ACK). As we only resend the selected damaged frames so we name this technique the Selective Repeat ARQ technique. The ACK and the NACK have the sequence number of the frame that helps the sender to identify the lost frame.

Important Points regarding the Selective Repeat ARQ:

- In the selective repeat ARQ, we only resend the data frames that are damaged or lost.
- If any frame is lost or damaged then the receiver sends a negative acknowledgment (NACK) to the sender and if the frame is correctly received, it sends back an acknowledgment (ACK).
- The sender sets a timer for each frame so whenever the timer is over and the sender has not received any acknowledgment, then the sender knows that the particular frame is either lost or damaged.
- As the sender needs to wait for the timer to expire before retransmission. So, we use negative acknowledgment or NACK.
- The ACK and the NACK have the sequence number of the frame that helps the sender to identify the lost frame.
- The receiver has the capability of sorting the frames present in the memory buffer using the sequence numbers.
- The sender must be capable enough to search for the lost frame for which the NACK has been received.
- The size of the sender's window is $2(\square-1)$2($m-1$), where m is the number of bits used in the header of the packet to express the packet's sequence number.
- The window size of the receiver is the same as that of the sender i.e. $2(\square-1)$2($m-1$).
- In Selective Repeat protocol, the efficiency is N/(1+2xa) where a is ratio of propagation delay vs the transmission delay and N is number of packets sent.
- It is efficient as only the lost or broken packets need retransmission.

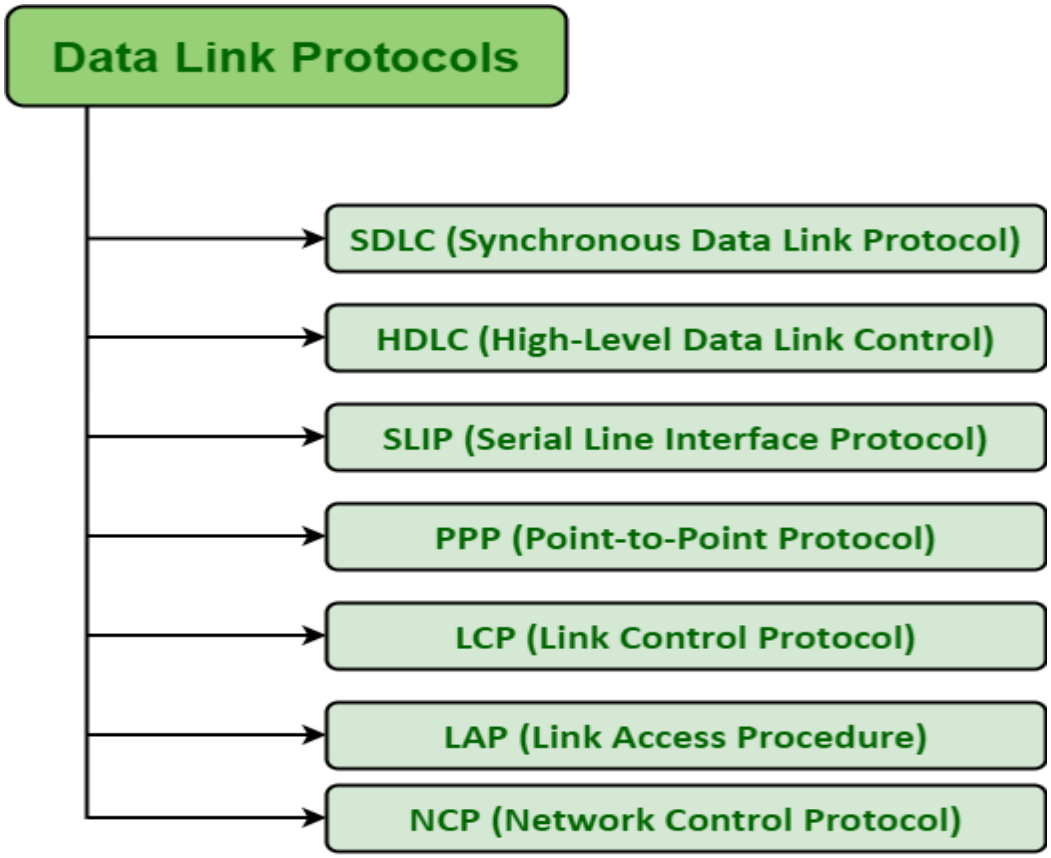
Let us briefly discuss ARQ. ARQ stands for Automatic Repeat Request. ARQ is an error-control strategy that ensures that a sequence of information is delivered in order and without any errors or duplications despite transmission errors and losses.

DATA LINK LAYER PROTOCOLS EXAMPLE
LECTURE 5

Data Link Layer protocols are generally responsible to simply ensure and confirm that the bits and bytes that are received are identical to the bits and bytes being transferred. It is basically a set of specifications that are used for implementation of data link layer just above the physical layer of the Open System Interconnections (OSI) Model.

Some Common Data Link Protocols :

There are various data link protocols that are required for Wide Area Network (WAN) and modem connections. Logical Link Control (LLC) is a data link protocol of Local Area Network (LAN). Some of data link protocols are given below :



1. **Synchronous Data Link Protocol (SDLC) –** SDLC is basically a communication protocol of computer. It usually supports multipoint links even error recovery or error correction also. It is usually used to carry SNA (Systems Network Architecture) traffic and is present precursor to HDLC. It is also designed and developed by IBM in 1975. It is also used to connect all of the remote devices to mainframe computers at central locations may be in point-to-point (one-to-one) or point-to-multipoint (one-to-many) connections. It is also used to make sure that the data units should arrive correctly and with right flow from one network point to next network point.

2. **High-Level Data Link Protocol (HDLC) –** HDLC is basically a protocol that is now assumed to be an umbrella under which many

Wide Area protocols sit. It is also adopted as a part of X.25 network. It was originally created and developed by ISO in 1979. This protocol is generally based on SDLC. It also provides best-effort unreliable service and also reliable service. HDLC is a bit-oriented protocol that is applicable for point-to-point and multipoint communications both.

3. **Serial Line Interface Protocol (SLIP) –** SLIP is generally an older protocol that is just used to add a framing byte at end of IP packet. It is basically a data link control facility that is required for transferring IP packets usually among Internet Service Providers (ISP) and a home user over a dial-up link. It is an encapsulation of the TCP/IP especially designed to work with over serial ports and several router connections simply for communication. It is some limitations like it does not provide mechanisms such as error correction or error detection.
4. **Point to Point Protocol (PPP) –** PPP is a protocol that is basically used to provide same functionality as SLIP. It is most robust protocol that is used to transport other types of packets also along with IP Packets. It can also be required for dial-up and leased router-router lines. It basically provides framing method to describe frames. It is a character-oriented protocol that is also used for error detection. It is also used to provides two protocols i.e. NCP and LCP. LCP is used for bringing lines up, negotiation of options, bringing them down whereas NCP is used for negotiating network-layer protocols. It is required for same serial interfaces like that of HDLC.

5. **Link Control Protocol (LCP) –** It was originally developed and created by IEEE 802.2. It is also used to provide HDLC style services on LAN (Local Area Network). LCP is basically a PPP protocol that is used for establishing, configuring, testing, maintenance, and ending or terminating links for transmission of data frames.

6. **Link Access Procedure (LAP) –** LAP protocols are basically a data link layer protocols that are required for framing and transferring data across point-to-point links. It also includes some reliability service features. There are basically three types of LAP i.e. LAPB (Link Access Procedure Balanced), LAPD (Link Access Procedure D-Channel), and LAPF (Link Access Procedure Frame-Mode Bearer Services). It is actually originated from IBM SDLC, which is being submitted by IBM to the ISP simply for standardization.

7. **Network Control Protocol (NCP) –** NCP was also an older protocol that was implemented by ARPANET. It basically allows users to have access to use computers and some of the devices at remote locations and also to transfer files among two or more computers. It is generally a set of protocols that is forming a part of PPP. NCP is always available for each and every higher-layer protocol that is supported by PPP. NCP was replaced by TCP/IP in the 1980s

MEDIUM ACCESS SUB LAYER

LECTURE 6

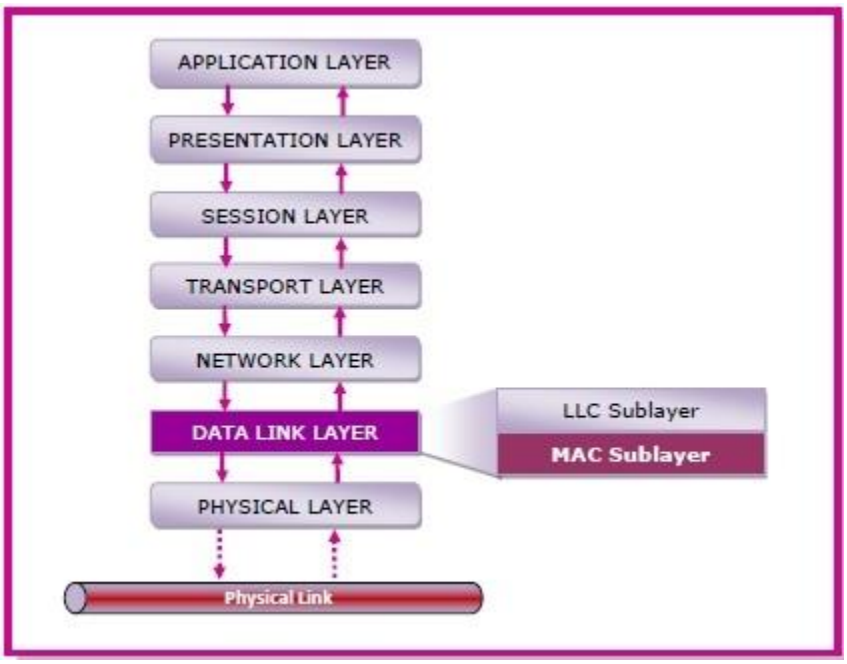
The medium access control (MAC) is a sublayer of the data link layer of the open system interconnections (OSI) reference model for data transmission. It is responsible for flow control and multiplexing for transmission medium. It controls the transmission of data packets via remotely shared channels. It sends data over the network interface card.

MAC Layer in the OSI Model

The Open System Interconnections (OSI) model is a layered networking framework that conceptualizes how communications should be done between heterogeneous systems. The data link layer is the second lowest layer. It is divided into two sublayers −

- The logical link control (LLC) sublayer
- The medium access control (MAC) sublayer

The following diagram depicts the position of the MAC layer −



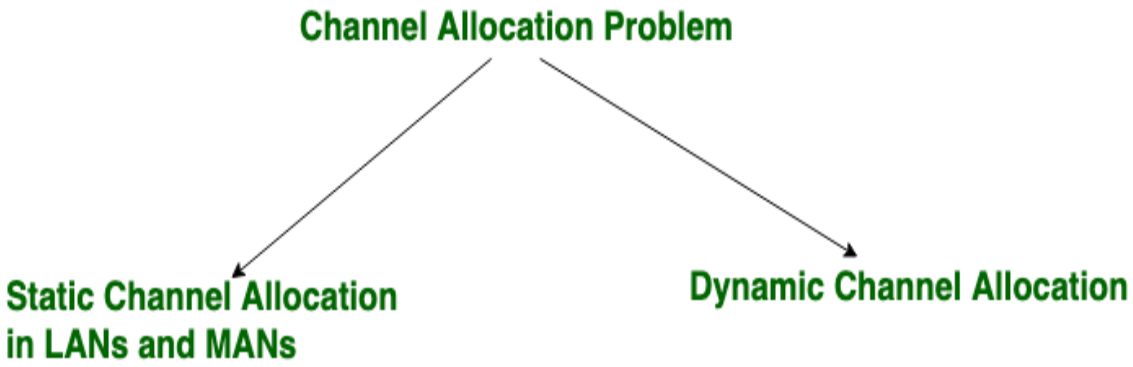
Functions of MAC Layer

- It provides an abstraction of the physical layer to the LLC and upper layers of the OSI network.
- It is responsible for encapsulating frames so that they are suitable for transmission via the physical medium.
- It resolves the addressing of source station as well as the destination station, or groups of destination stations.
- It performs multiple access resolutions when more than one data frame is to be transmitted. It determines the channel access methods for transmission.

- It also performs collision resolution and initiating retransmission in case of collisions.
- It generates the frame check sequences and thus contributes to protection against transmission errors.

Channel allocation: It is a process in which a single channel is divided and allotted to multiple users in order to carry user specific tasks. There are user's quantity may vary every time the process takes place. If there are N number of users and channel is divided into N equal-sized sub channels, Each user is assigned one portion. If the number of users are small and don't vary at times, then Frequency Division Multiplexing can be used as it is a simple and efficient channel bandwidth allocating technique.

- Channel allocation problem can be solved by two schemes: Static Channel Allocation in LANs and MANs, and Dynamic Channel Allocation.



These are explained as following below.

1. Static Channel Allocation in LANs and MANs:
It is the classical or traditional approach of allocating a single channel among multiple competing users using Frequency Division Multiplexing (FDM). if there are N users, the frequency channel is divided into N equal sized portions (bandwidth), each user being assigned one portion. since each user has a private frequency band, there is no interference between users.
It is not efficient to divide into fixed number of chunks.

T = 1/(U*C-L)

T(FDM) = N*T(1/U(C/N)-L/N)
Where,

T = mean time delay,
C = capacity of channel,
L = arrival rate of frames,
1/U = bits/frame,
N = number of sub channels,
T(FDM) = Frequency Division Multiplexing Time

2.DynamicChannelAllocation:
Possibleassumptionsinclude:

1. **StationModel:**
   Assumes that each of N stations independently produce frames. The probability of producing a packet in the interval IDt where I is the constant arrival rate of new frames.

2. **SingleChannelAssumption:**
   In this allocation all stations are equivalent and can send and receive on that channel.

3. **CollisionAssumption:**
   If two frames overlap in time-wise, then that's collision. Any collision is an error, and both frames must re transmitted. Collisions are only possible error.

4. **Time** can be divided into Slotted or Continuous.

5. **Stations** can sense a channel is busy before they try it.

ProtocolAssumption:

- N independent stations.
- A station is blocked until its generated frame is transmitted.
- probability of a frame being generated in a period of length Dt is IDt where I is the arrival rate of frames.
- Only a single Channel available.
- Time can be either: Continuous or slotted.
- **Carrier Sense:** A station can sense if a channel is already busy before transmission.
- **No Carrier Sense:** Time out used to sense loss data.

MULTIPLE ACCESS PROTOCOLS

LECTURE 7

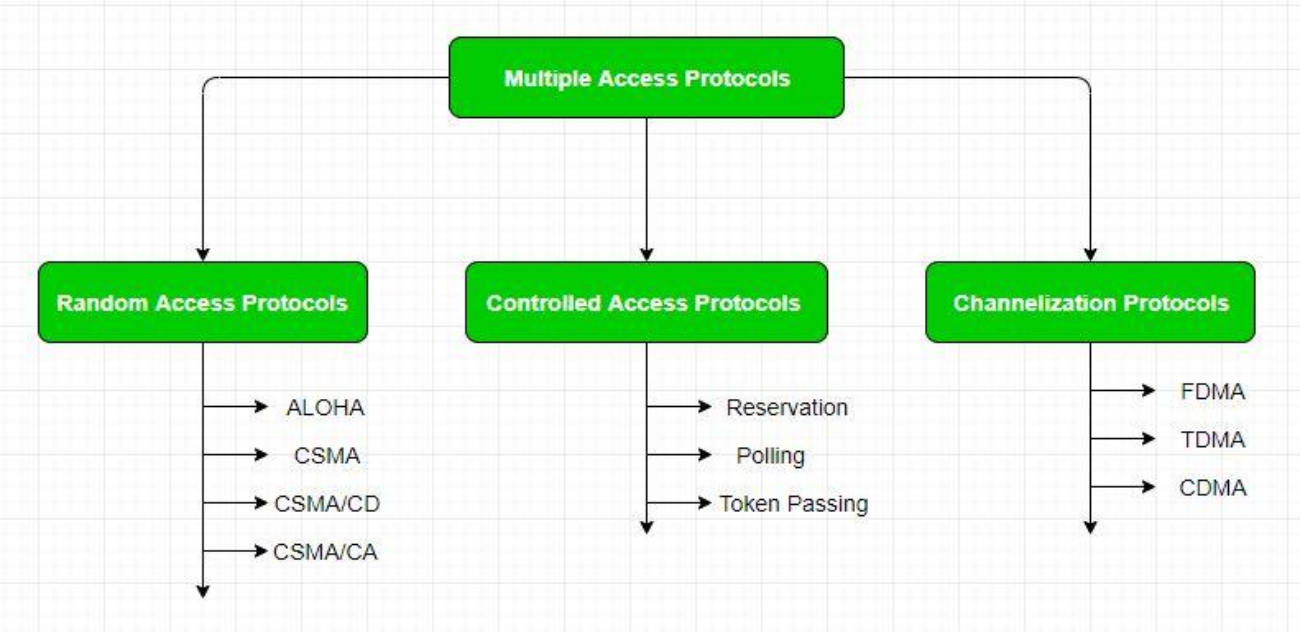
- The Data Link Layer is responsible for transmission of data between two nodes. Its main functions are-
  - Data Link Control
  - Multiple Access Control



DataLinkcontrol – The data link control is responsible for reliable transmission of message over transmission channel by using techniques like framing, error control and flow control.

MultipleAccessControl – If there is a dedicated link between the sender and the receiver then data link control layer is sufficient, however if there is no dedicated link present then multiple stations can access the channel simultaneously. Hence multiple access protocols are required to decrease collision and avoid crosstalk. For example, in a classroom full of students, when a teacher asks a question and all the students (or stations) start answering simultaneously (send data at same time) then a lot of chaos is created(data overlap or data lost) then it is the job of the teacher (multiple access protocols) to manage the students and make them answer one at a time.

Thus, protocols are required for sharing data on non dedicated channels. Multiple access protocols can be subdivided further as –



1. Random Access Protocol: In this, all stations have same superiority that is no station has more priority than another station. Any station can send data depending on medium's state(idle or busy). It has two features:

1. There is no fixed time for sending data
2. There is no fixed sequence of stations sending data

The Random access protocols are further subdivided as:

(a) ALOHA – It was designed for wireless LAN but is also applicable for shared medium. In this, multiple stations can transmit data at the same time and can hence lead to collision and data being garbled.

- **PureAloha:**
  When a station sends data it waits for an acknowledgement. If the acknowledgement doesn't come within the allotted time then the station waits for a random amount of time called back-off time (Tb) and re-sends the data. Since different stations wait for different amount of time, the probability of further collision decreases.

Vulnerable Time = 2* Frame transmission time

Throughput = G exp{-2*G}

Maximum throughput = 0.184 for G=0.5

- **SlottedAloha:**
  It is similar to pure aloha, except that we divide time into slots and sending of data is allowed only at the beginning of these slots. If a station misses out the allowed time, it must wait for the next slot. This reduces the probability of collision.

Vulnerable Time = Frame transmission time

Throughput = G exp{-*G}

Maximum throughput = 0.368 for G=1

For more information on ALOHA refer – LAN Technologies

(b) CSMA – Carrier Sense Multiple Access ensures fewer collisions as the station is required to first sense the medium (for idle or busy) before transmitting data. If it is idle then it sends data, otherwise it waits till the channel becomes idle. However there is still chance of collision in CSMA due to propagation delay. For example, if station A wants to send data, it will first sense the medium.If it finds the channel idle, it will start sending data. However, by the time the first bit of data is transmitted (delayed due to propagation delay) from station A, if station B requests to send data and senses the medium it will also find it idle and will also send data. This will result in collision of data from station A and B.

CSMA access modes-

- **1-persistent:** The node senses the channel, if idle it sends the data, otherwise it continuously keeps on checking the medium for being idle and transmits unconditionally(with 1 probability) as soon as the channel gets idle.
- **Non-Persistent:** The node senses the channel, if idle it sends the data, otherwise it checks the medium after a random amount of time (not continuously) and transmits when found idle.
- **P-persistent:** The node senses the medium, if idle it sends the data with p probability. If the data is not transmitted ((1-p) probability) then it waits for some time and checks the medium again, now if it is found idle then it send with p probability. This repeat continues until the frame is sent. It is used in Wifi and packet radio systems.
- **O-persistent:** Superiority of nodes is decided beforehand and transmission occurs in that order. If the medium is idle, node waits for its time slot to send data.

(c) CSMA/CD – Carrier sense multiple access with collision detection. Stations can terminate transmission of data if collision is detected. For more details refer – Efficiency of CSMA/CD

(d) CSMA/CA – Carrier sense multiple access with collision avoidance. The process of collisions detection involves sender receiving acknowledgement signals. If there is just one signal(its own) then the data is successfully sent but if there are two signals(its own and the one with which it has collided) then it means a collision has occurred. To distinguish between these two cases, collision must have a lot of impact on received signal. However it is not so in wired networks, so CSMA/CA is used in this case.

CSMA/CA avoids collision by:

1. **Interframe space –** Station waits for medium to become idle and if found idle it does not immediately send data (to avoid collision due to propagation delay) rather it waits for a period of time called Interframe space or IFS. After this time it again checks the medium for being idle. The IFS duration depends on the priority of station.
2. **Contention Window –** It is the amount of time divided into slots. If the sender is ready to send data, it chooses a random number of slots as wait time which doubles every time medium is not found idle. If the medium is found busy it does not restart the entire process, rather it restarts the timer when the channel is found idle again.
3. **Acknowledgement –** The sender re-transmits the data if acknowledgement is not received before time-out.

2.ControlledAccess:

In this, the data is sent by that station which is approved by all other stations.

3.Channelization:

In this, the available bandwidth of the link is shared in time, frequency and code to multiple stations to access channel simultaneously.

- **Frequency Division Multiple Access (FDMA) –** The available bandwidth is divided into equal bands so that each station can be allocated its own band. Guard bands are also added so that no two bands overlap to avoid crosstalk and noise.
- **Time Division Multiple Access (TDMA) –** In this, the bandwidth is shared between multiple stations. To avoid collision time is divided into slots and stations are allotted these slots to transmit data. However there is a overhead of synchronization as each station needs to know its time slot. This is resolved by adding synchronization bits to each slot. Another issue with TDMA is propagation delay which is resolved by addition of guard bands.

- **Code Division Multiple Access (CDMA) –** One channel carries all transmissions simultaneously. There is neither division of bandwidth nor division of time. For example, if there are many people in a room all speaking at the same time, then also perfect reception of data is possible if only two person speak the same language. Similarly, data from different stations can be transmitted simultaneously in different code languages.
- **Orthogonal Frequency Division Multiple Access (OFDMA) –** In OFDMA the available bandwidth is divided into small subcarriers in order to increase the overall performance, Now the data is transmitted through these small subcarriers. it is widely used in the 5G technology

LECTURE 8

WIRELESS LAN:

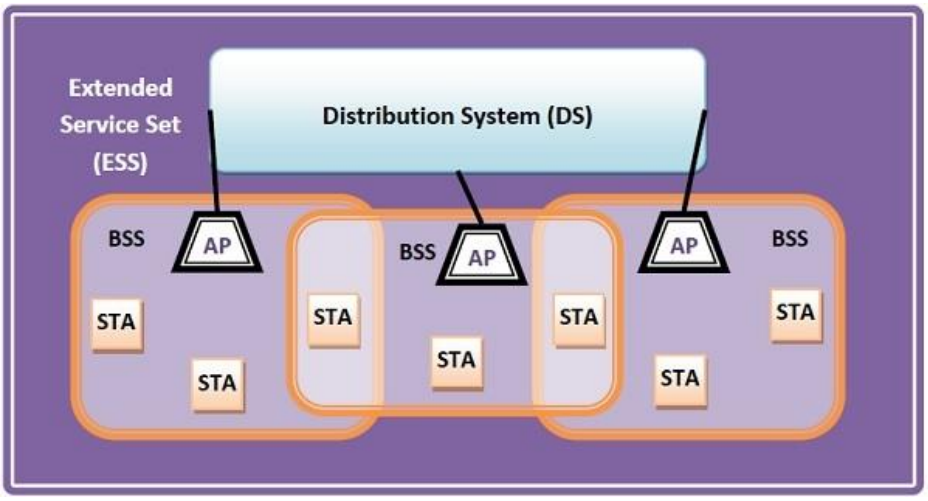
Wireless LANs (WLANs) are wireless computer networks that use high-frequency radio waves instead of cables for connecting the devices within a limited area forming LAN (**Local Area Network**). Users connected by wireless LANs can move around within this limited area such as home, school, campus, office building, railway platform, etc.

Most WLANs are based upon the standard IEEE 802.11 standard or WiFi.

Components of WLANs

The components of WLAN architecture as laid down in IEEE 802.11 are −

- **Stations (STA)** − Stations comprises of all devices and equipment that are connected to the wireless LAN. Each station has a wireless network interface controller. A station can be of two types −
  - Wireless Access Point (WAP or AP)
  - Client
- **Basic Service Set (BSS)** − A basic service set is a group of stations communicating at the physical layer level. BSS can be of two categories −
  - Infrastructure BSS
  - Independent BSS
- **Extended Service Set (ESS)** − It is a set of all connected BSS.
- **Distribution System (DS)** − It connects access points in ESS.



Types of WLANS:

WLANs, as standardized by IEEE 802.11, operates in two basic modes, infrastructure, and ad hoc mode.

- **Infrastructure Mode** − Mobile devices or clients connect to an access point (AP) that in turn connects via a bridge to the LAN or Internet. The client transmits frames to other clients via the AP.
- **Ad Hoc Mode** − Clients transmit frames directly to each other in a peer-to-peer fashion.

Advantages of WLANs

- They provide clutter-free homes, offices and other networked places.
- The LANs are scalable in nature, i.e. devices may be added or removed from the network at greater ease than wired LANs.
- The system is portable within the network coverage. Access to the network is not bounded by the length of the cables.
- Installation and setup are much easier than wired counterparts.
- The equipment and setup costs are reduced.

Disadvantages of WLANs

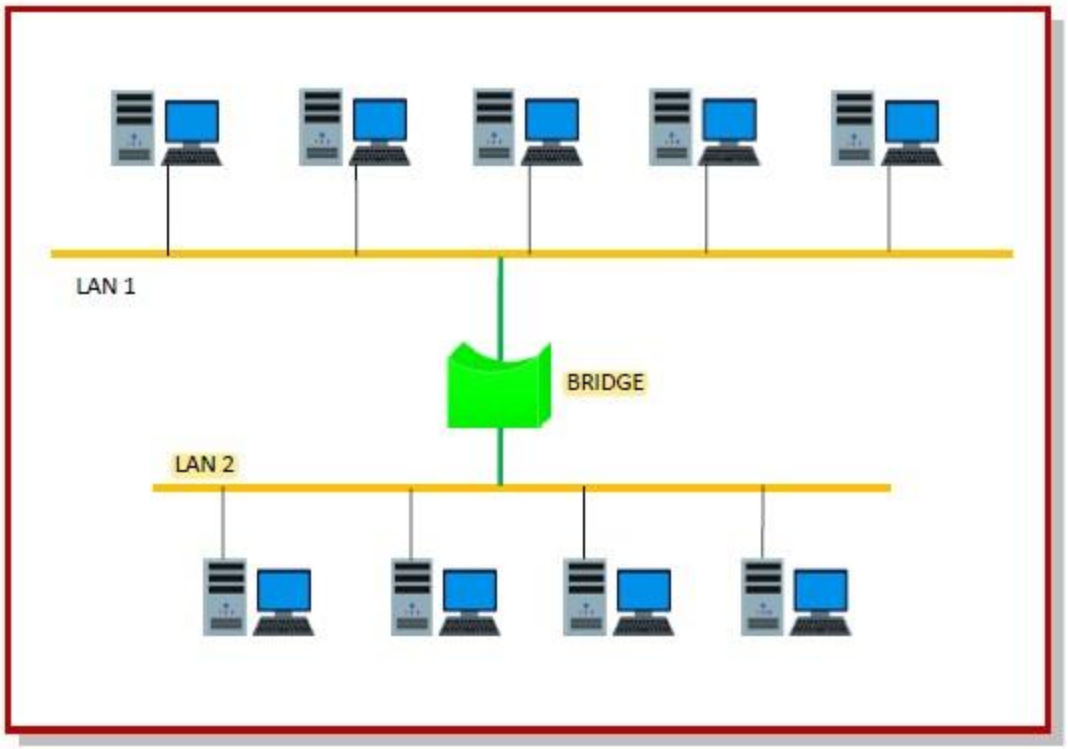
- Since radio waves are used for communications, the signals are noisier with more interference from nearby systems.
- Greater care is needed for encrypting information. Also, they are more prone to errors. So, they require greater bandwidth than the wired LANs.
- WLANs are slower than wired LANs.

Network switching is the process of forwarding data frames or packets from one port to another leading to data transmission from source to destination. Data link layer is the second layer of the Open System Interconnections (OSI) model whose function is to divide the stream of bits from physical layer into data frames and transmit the frames according to switching requirements. Switching in data link layer is done by network devices called **bridges**.

BRIDGES:

A data link layer bridge connects multiple LANs (local area networks) together to form a larger LAN. This process of aggregating networks is called network bridging. A bridge connects the different components so that they appear as parts of a single network.

The following diagram shows connection by a bridge −



SWITCHING BY BRIDGES:

When a data frame arrives at a particular port of a bridge, the bridge examines the frame's data link address, or more specifically, the MAC address. If the destination address as well as the required switching is valid, the bridge sends the frame to the destined port. Otherwise, the frame is discarded.

The bridge is not responsible for end to end data transfer. It is concerned with transmitting the data frame from one hop to the next. Hence, they do not examine the payload field of the frame. Due to this, they can help in switching any kind of packets from the network layer above.

Bridges also connect virtual LANs (VLANs) to make a larger VLAN.

If any segment of the bridged network is wireless, a wireless bridge is used to perform the switching.

There are three main ways for bridging −

- simple bridging
- multi-port bridging
- learning or transparent bridging

ETHERNET BRIDGES:

Ethernet Bridges are wireless radios that can be used to extend a wireless network to an Ethernet switch or hub (which can be used to extend connectivity to multiple wired

devices). Ethernet bridges can also be used to connect any device with an Ethernet port such as a Tivo, Xbox, or even a computer to the wireless network without having to install drivers or client software. This is a great solution for use with Mac OS and Linux computers, where drivers may be limited and more difficult to find.

Another benefit of using a wireless bridge is that since it uses wired Ethernet to deliver bandwidth to the client, you can extend the cat5 cable to its maximum segment length of 100 meters and still get connectivity. In theory, by using a Power over Ethernet (PoE) injector, you can send power over the Ethernet data cable as well and place the bridge as far away as 328 feet.

Most Ethernet bridges support external antennas. Figure 5.7 shows a Linksys WET11 with a removable RP-TNC antenna.